

MedAxis × Meridian — Data Science Handover Notes

■■ **HANDOVER STATUS: INCOMPLETE** — These notes were written by the outgoing DS under time pressure. Stages 4–12 are NOT documented. You will need to fill the gaps using the data files, company brief, and stakeholder conversations.

Stage 1 — Data Ingestion & Warehouse

We receive 8 source files from Meridian's EPR system (System C Careflow). The files land in an SFTP folder daily at 03:00 UTC. The ingestion pipeline loads them into PostgreSQL. Schema is roughly FHIR-aligned but not strict — see data dictionary.

Known issues: DISCHARGE_DATE is null for ~1.5% of rows (patients still admitted or data error — CHECK THIS). The PREV_ADM_90 column was added in March 2023 — earlier rows may have NULL. LOS_DAYS is calculated at ingest but sometimes disagrees with ADMISSION_DATE/DISCHARGE_DATE arithmetic — use the date columns, not the pre-calculated field.

- patients.csv: 2000 patients, 8 columns
- admissions.csv: primary fact table — all encounter-level data
- lab_results.csv: one row per test per encounter
- observations.csv: vital signs — multiple per encounter
- medications.csv: medication records per encounter
- clinical_notes_sample.csv: 200 discharge summaries (sample only)
- weekly_demand_forecast.csv: 2 years of weekly demand data

Stage 2 — Feature Engineering (partial)

The following features were partially implemented in a Jupyter notebook (not yet in a pipeline). The notebook is in the old repo — NOT DOCUMENTED HERE. You will need to re-implement from scratch using the feature list in the company brief.

- age_band: calculated from DATE_OF_BIRTH — WORKING
- comorb_count: direct from COMORB_CNT — WORKING
- prev_adm_90: direct from PREV_ADM_90 — CHECK NULL HANDLING
- emergency_flag: derived from ADM_TYPE=='EM' — WORKING
- abnormal_lab_flag: aggregated from lab_results — NOT COMPLETED
- medication_count: join to medications — NOT COMPLETED
- discharge_delay_flag: direct from DD_FLAG — DEPRECATED? Check with James
- days_since_last_adm: NOT IMPLEMENTED — needs window function

Stage 3 — Contacts

Key contacts for this project:

- Dr Priya Nair (Chief Medical Officer, Meridian) — clinical sign-off, model safety
- James Okafor (Head of Data Engineering, MedAxis) — data pipeline issues
- Sophia Reeves (Head of AI Governance, MedAxis) — responsible AI, model card
- Marcus Webb (Commercial Director, MedAxis) — contract, ROI, board prep

Open Questions — NOT RESOLVED

- What is the correct target variable for readmission? 30DR column or re-calculate from DISCHARGE_DATE + next ADMISSION_DATE? — ASK JAMES
- Week 68 in weekly_demand_forecast.csv shows anomalously low admissions. Was there a data pipeline failure that week? — NOT CONFIRMED
- ETHNICITY field: 'Unknown' vs blank — are these the same? Check with Meridian
- The SITE_ID MER005 has very low volume — is this site still active? — NOT DOCUMENTED
- ICD10_SEC field uses pipe separator — is this consistent across all years? — NOT VERIFIED

■■ Stages 4–12 NOT DOCUMENTED. You are responsible for all modelling, deployment, and governance phases.